SANTHOSH KUMAR
AI/ML Engineer | Python Developer | Machine Learning Enthusiast
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PROFESSIONAL SUMMARY

Aspiring AI/ML Engineer pursuing a B.E. in Computer Science and Engineering with specialization in Artificial Intelligence & Machine Learning. Strong foundation in Python, Data Structures & Algorithms, Machine Learning, Deep Learning, NLP and Generative AI. Experienced in academic and personal projects involving predictive analytics, computer vision, NLP and AI-powered applications.
Passionate about building practical intelligent systems and solving real-world problems through AI and software engineering.

TECHNICAL SKILLS

Programming: Python, Java, C, JavaScript, SQL,

AI / ML: Machine Learning, Deep Learning, NLP, Computer Vision, Predictive Analytics

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow/Keras, PyTorch, OpenCV, XGBoost, Matplotlib

Web / Backend: React, FastAPI, Flask, REST APIs, HTML5, CSS3

Databases: MySQL, MongoDB, PostgreSQL

Tools: Git, GitHub, Docker, VS Code, Linux

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

NAYAN-X — Predictive, Memory-Augmented Collision Avoidance for Unstructured Indian Roads

Technologies: Python, Computer Vision, Deep Learning, Sensor Fusion, AI/ML

- Designed an AI-driven collision-avoidance architecture for autonomous vehicles on unstructured Indian roads.
- Designed the SEE □ PREDICT □ REMEMBER □ ASSESS □ ADAPT □ AVOID □ LEARN system concept for predictive safety.
- Integrated perception, tracking, motion/intent prediction, risk assessment, adaptive planning and safety supervision concepts.

GitHub: github.com/SanthoshKumar0710/NAYAN-X

AI Infrastructure Failure Prediction Platform

Technologies: Python, Machine Learning, FastAPI, React, PostgreSQL, Redis, Docker

- Developed an AI-based platform to analyze infrastructure metrics and predict potential failures.
- Implemented data processing, anomaly detection, predictive analysis and risk assessment workflows.
- Built FastAPI backend services and React frontend components; used Docker for service orchestration.

GitHub: github.com/SanthoshKumar0710/ai-infra-failure-prediction

Lead Scoring Prediction

Technologies: Python, Pandas, Scikit-learn, Machine Learning, Data Visualization

- Developed a machine learning model to predict lead conversion probability.
- Performed data preprocessing, feature analysis and classification model development.
- Generated predictive insights to support prioritization of high-potential leads.

Learning Rate Analysis in Deep Neural Network Training

Technologies: Python, TensorFlow/Keras, NumPy, Pandas, Matplotlib, Scikit-learn

- Analyzed how different learning rates affect neural-network training and convergence.
- Compared training and validation performance across learning-rate configurations.
- Visualized loss and accuracy trends to evaluate model-training behavior.

GitHub: github.com/Santhoshkumar/DL-MINI-PROJECT

AuraMart — Enterprise Full-Stack E-Commerce Platform

Technologies: Python, FastAPI, React 19, TypeScript, Vite, Tailwind CSS v4, PostgreSQL, Redis, SQLAlchemy, Docker

- Developed a full-stack e-commerce platform with a modern, responsive React frontend and a FastAPI REST API backend.

- Implemented relational database management, Redis-based caching and session management, and secure payment processing with Razorpay signature verification and Cash on Delivery (COD).
- Integrated product image and file upload handling with MIME-type validation and containerized services using Docker for streamlined deployment.

GitHub:https://github.com/SanthoshKumar0710/e-commerce.git

EDUCATION

B.E. Computer Science and Engineering — Artificial Intelligence & Machine Learning

Rajalakshmi Institute of Technology, Chennai, Tamil Nadu | Expected Graduation: 2028

Higher Secondary Education — Don Bosco Matric Higher Secondary School, Chennai | Completed: 2024

CERTIFICATIONS

- Google AI Essentials
- Machine Learning for Data Science — Project Certificate
- UI PATH Certifications
- MongoDB
- RAG Certification
- Machine Learning Foundations -AWS
- SQL Certifications
- Gen

RELEVANT COURSEWORK

Data Structures & Algorithms, Machine Learning, Deep Learning, Data Analytics, Artificial Intelligence, Database Management Systems, Operating Systems, Computer Networks